STAT5351 Lecture Notes

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1 Lecture 1 - 24/08/26

1.1 Counting principle

Suppose you can solve a problem in 2 steps, with n_1 ways for step 1 and n_2 ways for step 2. How many ways can you solve the problem?

$$\begin{matrix} (x_1, y_1) & (x_1, y_2) & \dots & (x_1, y_{n_2}) \\ \vdots & & & \\ (x_{n_1}, y_1) & (x_{n_1}, y_2) & \dots & (x_{n_1}, y_{n_2}) \end{matrix}$$

Thus, there are $n_1 \cdot n_2$ ways.

Theorem 1.1. *Given r steps to solve a problem and n_i ways to solve step i, then the problem can be solve in $n_1 \cdot n_2 \cdot \dots \cdot n_r$ ways.*

1.2 Permutations

Definition 1.2. A permutation on n different elements is a **DISTINCT** arrangement of the n elements, where distinct arrangement means no repetition of elements in the arrangement.

Theorem 1.3. *The number of permutations of n unique elements is $n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 2 \cdot 1 = n!$.*

Theorem 1.4. *The number of permutations of n elements taken r at a time is* $nPr = n(n-1)(n-2)\cdots(n-r+1) = \frac{n!}{(n-r)!}$

Theorem 1.5. *Suppose n elements have k different types: n_1 of type 1 to n_k of type k. Then $n = n_1 + \cdots + n_k$ and the number of permutations of n elements is $\frac{n!}{n_1!n_2!\cdots n_k!}$.*

1.3 Combinations

Definition 1.6. A combination of n different elements taken r at a time is a selection of r elements from n elements without regard to the order of selection.

Theorem 1.7. *The number of permutations of n elements taken r at a time (n choose r) is*

$$nPr = \frac{n!}{(n-r)!r!}$$

2 Lecture 2 - 26/08/26

2.1 Partitions

Definition 2.1. A partition of n elements into k subsets is a way to separate n elements into k groups without regard for order.

Theorem 2.2. *The number of ways in which n elements can be partitioned into k subsets such that n_1 elements in subset 1, n_2 in subset 2, etc is*

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

3 Lecture 3 - 31/08/26

3.1 Discrete probability function

Discrete probability function P has the following properties:

1. $P(S) = 1$.
2. $P(\emptyset) = 0$.
3. $P(A^c) = 1 - P(A)$.
4. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
5. If $A \cap B = \emptyset$, $P(A \cup B) = P(A) + P(B)$.
6. $A, B \subset S$, $A \subseteq B \implies P(A) \leq P(B)$.

4 Lecture 4 - 02/09/26

4.1 Conditional Probability

If B has occurred, what is the probability of A?

Theorem 4.1. $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Theorem 4.2. $P(A \cap B) = P(A|B) \cdot P(B)$

Theorem 4.3. $P(A \cap B \cap C) = P(A)P(B|A)P(C|A \cap B)$

5 Lecture 5 - 09/09/26

5.1 Bayes' Theorem

Definition 5.1. $A_1, \ldots, A_k \subset B$ form a partition of B if A_i are all mutually exclusive.

Remark 5.2. For a partiton of B,

$$P(B) = P(A_1 \cap B) + \cdots + P(A_k \cap B) = P(B|A_1)P(A_1) \cdots P(B|A_k)P(A_k)$$

Theorem 5.3. $P(B) = \sum_{n=1}^{k} P(B|A_i)P(A_i)$.

Theorem 5.4 (Bayes' Theorem)**.**

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{\sum_j P(B|A_j)P(A_j)}$$

Bayesian inference is forming an event B to conclude the cause, i.e. which A_i is the cause of B.

5.2 Independent Events

Definition 5.5. Two events are independent if

$$P(A \cap B) = P(A)P(B)$$

if they are not independent they are dependent.

Theorem 5.6. *If A and B are independent, then A and B^c are independent.*

6 Lecture 6 - 14/09/26

6.1 Random Variables

Definition 6.1. Let S be a sample space with a probability function P. A random variable is a function $X : S \to \mathbb{R}$. The notation is $P(X = b) :=$ probability of event $\{X = b\}$.

7 Lecture 7 - 16/09/26

7.1 Probability mass function

Theorem 7.1. *Let X be a discrete random variable on S such that $b_1, \ldots, b_n$ are the values in the image of X. Then $\{X = b_1\}, \ldots, \{X = b_n\}$ form a partition of S.*

Definition 7.2. The probability mass function of X is

$$f(y) := P(X = y), \quad \forall y \in \text{image of } X$$

Theorem 7.3. *If $f(y)$ is a pmf of X, then*

1. $f(y) \geq 0$.
2. $\sum_y f(y) = 1 \iff \sum f(b_i) = 1$.

7.2 Cumulative Distribution

Definition 7.4. $F(y) := P(X \geq y), \quad \forall y \in -\infty < y < \infty$ is the cumulative distribution of X. It measures the accumulation of probability as y increases.

8 Lecture 8 - 21/09/26

Example 8.1. $F(y)$ is the cumulative distribution of X.

$$F(y) = \begin{cases} 0 & y < 1 \\ \frac{1}{3} & 1 \leq y < 4 \\ \frac{1}{2} & 1 \leq 4 < 6 \\ \frac{5}{6} & 6 \leq 4 < 10 \\ 1 & y \geq 10 \end{cases}$$

$P(X = 4) = ?$, $P(2 < X \leq 6) = ?$
First, find the probability mass function (p.m.f). X takes values $b_1 = 1, b_2 = 4, b_3 = 6, b_4 = 10$.
Recall that $F(b_{i+1}) = F(b_i) + f(b_i + 1)$ where f is the p.m.f.

$$f(b_1) = f(1) = F(1) - 0 = \frac{1}{3}$$

$$f(b_2) = f(4) = F(4) - F(1) = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

$$f(b_3) = \frac{1}{3}$$

$$f(b_4) = \frac{1}{6}$$

Thus,

$$P(X = 4) = f(4) = \frac{1}{6}$$

and since 4 is inside the interval $(2, 6]$, then

$$(X = 4) \cup (X = 6)$$

and since the events are mutually exclusive,

$$P(2 < X \leq 6) = P(X = 4) + P(X = 6)$$

$$= f(4) + f(6) = \frac{1}{2}$$

Example 8.2. X is a discrete random variable with cumulative distribution

$$F(x) = \begin{cases} 0 & y < -1 \\ \frac{1}{9} & -1 \leq y < 1 \\ \frac{5}{9} & 1 \leq y < 3 \\ \frac{8}{9} & 3 \leq y < 5 \\ 1 & y \geq 5 \end{cases}$$

Find the pmf of X and $P(0 \leq X < 5)$.
X takes values $-1, 1, 3, 5$. Find the pmf:

$$f(-1) = \frac{1}{9}$$

$$f(1) = \frac{5}{9} - \frac{1}{9} = \frac{4}{9}$$

$$f(3) = \frac{1}{3}$$

$$f(5) = \frac{1}{9}$$

$$P(0 \leq X < 5) = P(X = 1) + P(X = 3) = f(1) + f(3) = \frac{7}{9}$$

8.1 Continuous random variables

Definition 8.3. Let S be a continous sample space. $X : S \to \mathbb{R}$ is a continous random variable if there exists a function $f(t) \geq 0$ defined on $\mathbb{R}$ such that

$$P(a \leq X \leq b) = \int_a^b f(t)dt$$

$f(t)$ is called the probability density function of X.

Remark 8.4. $f(t) \geq 0$ is defined for ALL $t \in \mathbb{R}$.

Theorem 8.5. *Let X be a continuous random variable, and f the pdf of X. Then*

- $P(X = a) = 0$

- $P(a \leq X \leq b) = P(a \leq X < b) = \ldots$
- $\int_{-\infty}^{\infty} f(t)\,dt = 1$

Definition 8.6. Let X be a continous random variable and $f(t)$ the pdf. THen

$$F(y) = P(X \leq y) = \int_{-\infty}^{y} f(t)\,dt$$

is called the cumulative distribution function of X.

Theorem 8.7. *Let f be the pdf and F the cdf of X.*

- $P(a \leq X \leq b) = F(b) = F(a)$
- *if f is a continuous function on $\mathbb{R}$ then $f(y) = \frac{dF(y)}{dy}$.*

8.2 Comparison of discrete and continous random variables X

8.2.1 Discrete

- f is pmf
- f defined on discrete values
- F is a cumulative distribution
- $F(y) = \sum_{b_i \leq y}^{\infty} f(b_i)$
- $P(a < X \leq B) = F(b) - F(a)$, the endpoints matter

8.2.2 Continuous

- f is a pdf
- f is defined on $\mathbb{R}$
- F is a cdf
- $F(y) = \int_{-\infty}^{y} f(t)\,dt$
- $P(a \leq X \leq b) = \int_{a}^{b} f(t)\,dt$, the endpoints don't matter.

9 Lecture 9 - 23/09/26

9.1 Joint/marginal distribution of discrete random variables

For discrete random variables X, Y, can we find $P(X = Y)$ by using the pmf f, g respectively? No, we need more information about their relation.

Definition 9.1. X, Y disc. rand. vars.

$$f(x, y) = P(X = x, Y = y)$$

for the pair (x, y) in the range of X, Y is called the joint probability mass function of X, Y.

Theorem 9.2. *if $f(x, y)$ is joint pmf,*

1. $f(x, y) > 0$
2. $\sum_x \sum_y f(x, y) = 1$

Definition 9.3. Joint cumulative distribution of discrete random variables X, Y is defined as

$$F(x, y) = P(X \leq x, Y \leq y) = \sum_{s \leq x} \sum_{t \leq y} f(s, t)$$

Remark 9.4. Joint pmf is defined on discrete values but the joint cumulative distribution is defined on $\mathbb{R}^2$.

Suppose we are given $f(x, y)$, the joint pmf of X and Y. How do we find the pmfs of X and Y? Idk.

Definition 9.5. $f(x, y)$ the joint pmf of X, Y. For each x in range of X,

$$g(x) = \sum_y f(x, y)$$

is called the marginal pmf of X. The marginal pmf of X is just the pmf of X.

Definition 9.6. $f(x, y)$ is a joint pdf of X, Y. if it satisfies

1. $f(x, y) \geq 0$ it is defined for all $(x, y) \in \mathbb{R}^2$.
2. $P(X, Y) \in A = \int \int_A f(x, y) dx dy$

Theorem 9.7. *$f(x, y)$ is a joint pdf. Then*

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx dy = 1$$

10 Lecture 10 - 28/09/26

Exam 1 Review.

11 Lecture 11 - 05/10/26

11.1 Marginal pdfs and pmfs

Definition 11.1. Let X, Y be continuous random variables. The joint c.d.f.

$$F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(s, t)\, ds\, dt$$

where $f(x)$ is the joint p.d.f.

Question: Given the joint pmf of X, Y, can we find the pdf of X and Y?

Definition 11.2. X, Y continuous random variables. Let $f(x, y)$ be the joint pdf. The marginal pdf of X is

$$g(x) = \int_{-\infty}^{\infty} f(x, y)\, dy, \quad -\infty < x < \infty$$

Remark 11.3. The marginal pdfs are just the pdfs of X and Y respectively.

Definition 11.4. Let $f(x, y)$ be the joint pmf of discrete random variables X, Y. Let $h(y)$ be the marginal pmf of Y. Then

$$f(x|y) = P(X = x, Y = y) = \frac{f(x, y)}{h(y)}$$

for all $x \in \text{range}(x)$. $f(x|y)$ is called the conditional pmf of X given $Y = y$.

Remark 11.5. We can regard the conditional probability as a function of two variables.

12 Lecture 12 - 07/10/26

Definition 12.1. The joint pdf is $f(x, y)$ of continuous random variables X, Y. Let $h(y)$ be the marginal pdf of Y.

$$f(x|y) = \frac{f(x, y)}{h(y)}$$

is the conditional pdf of X given $Y = y$. We can regard the conditional pdfs as functions of two variables.

Definition 12.2. $f(x)$ is the joint pdf of continuous random variables. $g(x)$ is the marginal pdf of X. $h(y)$ marginal pdf for Y. Then X, Y are independent if

$$f(x, y) = g(x)h(y), \quad \forall x, y \text{ s.t. } -\infty < x < \infty, \ -\infty < y < \infty$$

Remark 12.3. If we are given the joint pmf/pdf, then we can find the marginal ones. The converse is not necessarily true. However, if we are given the pdf/pmfs of X, Y, then in general we can't find the joint pdf/pmf. When X, Y are independent this is a special case where we can.

12.1 Expectation

After doing some random event many many times, what is the expected value? The "average" of the random variable with respect to the probability distribution.

Definition 12.4. X is a discrete random variable. $f(x)$ is the pmf. The expectation of X is

$$E(X) = \sum_x x \cdot f(x) \in \mathbb{R}$$

where you sum over all possible values of X. For continuous random variable,

$$E(X) = \int_{-\infty}^{\infty} x f(x)\, dx \in \mathbb{R}$$

13 Lecture 13 - 12/10/26

13.1 Expectation value for composed functions

Definition 13.1. Let X be a random variable on S. Define a function

$$g(t) : \mathbb{R} \to \mathbb{R}$$

Then $g(X)$ is the composition of g and X. It is a new random variable on S.

Theorem 13.2. *Let X be a discrete random variable and $f(x)$ is the pmf. Then*

$$E(g(X)) = \sum_x g(x) \cdot f(x)$$

Theorem 13.3. *Let X be a continuous random variable and $f(x)$ is the pdf of X. Then*

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) f(x)\, dx$$

Theorem 13.4. *Let $a, b \in \mathbb{R}$ be constants. Then*

$$E(aX + b) = aE(x) + b$$

Theorem 13.5. *let $c_1, c_2, \ldots, c_n$ be constants and $g_i : \mathbb{R} \to \mathbb{R}$. Then*

$$E(\sum_{n=1}^{n} c_i g_i(X)) = \sum_{i}^{n} c_i E(g_i(X))$$

Theorem 13.6. *Let X, Y be discrete random variables and $f(x, y)$ is a joint pmf. then*

$$E(g(X, Y)) = \sum_x \sum_y g(x, y) f(x, y)$$

13.2 Moments, mean, standard deviation, variance

Definition 13.7. the rth moment about the origin of random variable X is $\mu'_r = E(X^r)$.

Remark 13.8. For $r = 0$, $\mu'_0 = E(1) = 1$.
For $r = 1$, $\mu'_1 = E(X)$ is called the mean of X, denoted by μ.

Definition 13.9. the rth moment about the mean denoted by μ_r is

$$\mu_r = E((X - \mu)^r)$$

where μ is the mean of X.

Definition 13.10. Let X be a continuous random variable and $f(x)$ the pdf. Then the mean is given by

$$\mu = \int_{-\infty}^{\infty} x f(x)\, dx$$

and

$$\mu_r = \int_{-\infty}^{\infty} (x - \mu)^r f(x)\, dx$$

Theorem 13.11. $\sigma^2 = \mu'_2 - \mu^2 = E(X^2) - E(X)^2$

Remark 13.12. $\sigma^2 = E((x - \mu)^2) = \sum_x (x - \mu)^2 f(x)$ for the discrete case and $\int_{-\infty}^{\infty} (x - \mu)^2 f(x)\, dx$ for the continuous case.

14 Lecture 14 - 14/10/26

Theorem 14.1. *Let X have variance σ^2. Consider $aX + b$ where a, b are constant. Then it has*

$$Var(aX + b) = a^2 \sigma^2$$

14.1 Intepretations of standard deviation

Remark 14.2. The standard deviation describes how far the values of X are spread from the mean. Consider the case when X is a discrete random variable.

$$\sigma^2 = E[(X-\mu)^2] = \sum (x-\mu)^2 f(x)$$

where $f(x)$ is the pmf. The pmf is a "weighting", so then the σ^2 measures the "weighted average" squared distance to the mean. Suppose that X is a continuous random variable, then

$$P(|x-\mu| \leq t) = P(-t \leq x-\mu \leq t) = P(-t+\mu \leq X \leq \mu+t) = \int_{\mu-t}^{\mu+t} f(x)\,dx$$

where $f(x)$ is the pdf. In practice, we may not have all the information about f to calculate the integral above. Since σ^2 describes the spread of the distribution we can estimate the probability using it. This motivates the following theorem:

Theorem 14.3 (Chebyshev's Inequality)**.**

$$P(|x-\mu| \leq t) \geq 1 - \frac{\sigma^2}{t^2}$$

14.2 Moment generating function

Definition 14.4. A moment generating function of a random variable X is

$$M_X(t) = E(e^{tX})$$

For a discrete random variable,

$$M_X(t) = \sum_x e^{tx} f(x)$$

For continuous,

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x)\,dx$$

Remark 14.5. The following properties of $M_X(t)$ hold

1. $M_X(t)$ is a function of t.
2. $M_X(t)$ may not be defined for all t.
3. If $M_X(t)$ is defined, $M_X(t) \geq 0$.

Consider the series expansion of e^{tx}.

$$e^{tx} = 1 + tx + \frac{t^2x^2}{2} + \frac{t^3x^3}{3!} + \cdots$$

Let X be a continuous random variable. then

$$\begin{aligned}
M_X(t) &= \int_{-\infty}^{\infty} (1 + tx + \frac{t^2x^2}{2} + \cdots) f(x)\,dx \\
&= \int_{-\infty}^{\infty} f(x)\,dx + t\int_{-\infty}^{\infty} xf(x)\,dx + \frac{t^2}{2!}\int_{-\infty}^{\infty} x^2 f(x)\,dx + \cdots \\
&= \mu_0' + \mu_1' t + \mu_2' \frac{t^2}{2!} + \cdots + \mu_r' \frac{t^r}{r!} + \cdots \\
&= \sum_{r=0}^{\infty} \mu_r' \frac{t^r}{r!}
\end{aligned}$$

Theorem 14.6.

$$\mu'_r = \frac{d^r M_X(t)}{dt^r}|_{t=0} = M_X^{(r)}(0)$$

Theorem 14.7. *Let a, b be constants. Then*

1. $M_{X+a}(t) = E(e^{(x+a)t}) = e^{at} M_X(t)$
2. $M_{bX}(t) = E(e^{bXt}) = M_X(bt)$
3. $M_{\frac{x+a}{b}}(t) =$

15 Lecture 15 - 19/10/26

15.1 Product moments

Definition 15.1. The rth and sth product moment about the origin of random variables X and Y denoted by $\mu'_{r,s}$ is

$$\mu'_{x,y} = E(X^r Y^s), \quad r, s = 0, 1, 2, 3, \ldots$$

Recall that $X^r Y^s$ is a random variable on S. and if X, Y discrete,

$$E(X^r Y^s) = \sum_x \sum_y x^r y^s f(x, y)$$

for $f(x, y)$ joint pmf.

Definition 15.2. rth and sth product moment about the means of X, Y is

$$u_{r,s} = E((X - \mu_X)^r (Y - \mu_Y)^s)$$

where μ_i are the means respectively.

Definition 15.3. $\mu_{1,1}$ is called the covariance of X and Y denoted by $C[X, Y]$.

Theorem 15.4.

$$C[X, Y] = E(XY) - E(X)E(Y) = \mu'_{1,1} - \mu_X \mu_Y$$

Remark 15.5. The covariance measures the joint variability of X and Y. If larger values of X correspond to larger values of Y, then $C[X, Y] > 0$. If larger values of X are more likely to correspond to smaller values of Y, then $C[X, Y] < 0$.

15.2 Expectation of independent variables

Theorem 15.6. *If X, Y are independent,*

$$E(XY) = E(X)E(Y), \quad C[X, Y] = 0$$

16 Lecture 16 - 21/10/26

Theorem 16.1. *If $X_1, X_2, \ldots$ are pairwise independent,*

$$Y = \sum_{i=1}^{n} a_i X_i$$

then

$$\sigma_Y^2 = \sum_{i=1}^{n} a_i^2 \sigma_{X_i}^2$$

And if Y_1, Y_2 defined above but for different constants,

$$\sigma_{Y_1,Y_2}^2 = \sum a_i b_i \sigma_{X_i}^2$$

Theorem 16.2. *Properties of covariance:*

1. $\sigma_X^2 = C[X, X]$
2. $C[aX, Y] = aC[X, Y]$
3. $C[X, Y] = C[Y, X]$
4. $C[X + Y, Z] = C[X, Z] + C[Y, Z]$

Theorem 16.3.

$$\sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2 + 2C[X, Y]$$
$$\sigma_{X-Y}^2 = \sigma_X^2 + \sigma_Y^2 - 2C[X, Y]$$

Definition 16.4. Correlation

$$P(X, Y) = \frac{C[X, Y]}{\sigma_X \sigma_Y}$$

16.1 Conditional expectation

Suppose X, Y are random variables and $Y = y$. What is the expetation/mean/variance of X? The answer is they are called the conditional expectation/mean/variance of X given $Y = y$.

Definition 16.5. let $u(X)$ be a function of X. then

$$E(u(x), |y)$$

is the conditional expectation of $u(x)$ given $Y = y$. For discrete,

$$E(u(x)|y) = \sum_x u(x) f(x|y)$$

where $f(x|y)$ is the conditional pmf given $Y = y$. Same for continuous except with $f(x|y)$ the conditional pdf.

Definition 16.6. The conditional mean of X given $Y = y$ is

$$\mu_{x|y} = E(X|y)$$

Definition 16.7. Conditional variance is

$$\sigma_{X|y}^2 = E((x - \mu_{X|y})^2|y)$$

Remark 16.8. IMPORTANT.

$$\sigma_{X|y}^2 = E(X^2|y) - E(X|y)^2$$

still holds for conditional variance.

16.2 Special discrete random variables

Let X be discrete and take values $x_1, x_2, \ldots, x_k$. Assume

$$P(X = x_1) = P(X = x_2) = \cdots = P(x = x_k)$$

Then the pmf is

$$f(x_i) = P(X = x_i)$$

and thus

$$\sum f(x_i) = 1 \implies f(x_i) = \frac{1}{k}$$

Definition 16.9. A discrete random variable X has a uniform distribution if

$$f(x) = \frac{1}{k}$$

where $f(x)$ is the pmf of X

Definition 16.10. Let X take values of either 0 or 1. Then X has a Bernoulli distribution if its pmf is

$$f(x, \theta) = \theta^x (1 - \theta)^{1-x}, \quad x = 0, 1$$

17 Lecture 17 - 26/10/26

Definition 17.1. A discrete random variable has a binomial distribution if its pmf is

$$b(x, n, \theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}$$

Theorem 17.2.

$$b(x, n, \theta) = b(n - x, n, 1 - \theta)$$

Notice that it is hard ot get the mean and variance of X with the definition of binomial distribution. Thus,

Theorem 17.3. *The moment generating function of the binomial distribution is*

$$M_X(t) = (1 + \theta(e^t - 1))^n$$

Theorem 17.4. *For the binomial distribution,*

$$\mu = n\theta, \quad \sigma^2 = n\theta(1 - \theta)$$

Remark 17.5. The Bernoulli distribution is a special case of the binomial distribution where $b(x, 1, \theta)$.

We want to further understand the binomial distribution. Our goal is to fix the mean and find an approxiamation for $b(x, n, \theta)$ where n is very large, i.e. $n \to \infty$.

Definition 17.6. For a discrete random variable X, it has a Poisson distribution if its pmf is

$$P(x, \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Theorem 17.7. *The moment generating function of the Poisson distribution is*

$$M_X(t) = e^{\lambda(e^t - 1)}$$